Wages vs Amenities in Mobility Decisions Later in Life

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Outline

Motivation

- Typically, location choice models focus on wages.
- Do other factors play a role in location decisions later in life?
- **Main Question:** How does motivation for moving change with age?

Data Source

- IPUMS USA data from 1990 - 2019
- Decennial censuses and ACS data
- CMS data for health care costs
- DOE data for average climate conditions
- Panel structure by linking individuals across years

Sample Selection

- Individuals aged 40-80
- Moved at least once during observation period
- Exclude group quarters, military, and missing data
- Final sample size: 1.25 million households

Empirical Motivation

- Younger movers and older movers exhibit different mobility patterns
- High income individuals move more with higher frequency
- Those without children move more often
- Probability of moving peaks at the age around retirement, and then declines

Summary Statistics

Table: Summary Statistics

	Mean
n	21,138,046
Age	57.62
Wages	33251.83
Gross Rent	170.07
Health Costs	6336.11
Mover %	17%

Graphical Evidence

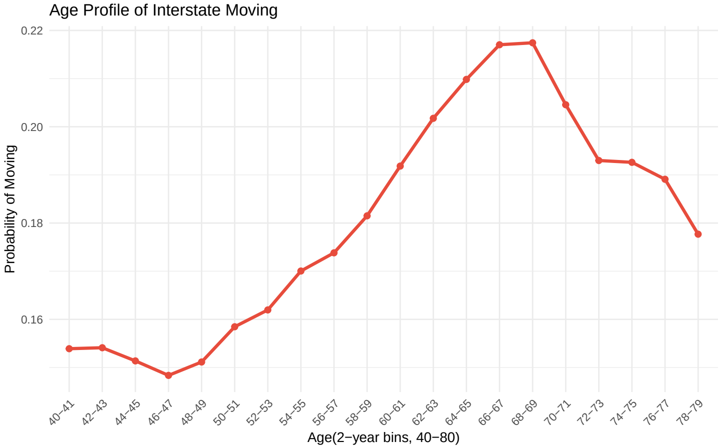

Figure: Age Profile of Moving Frequency

Graphical Evidence

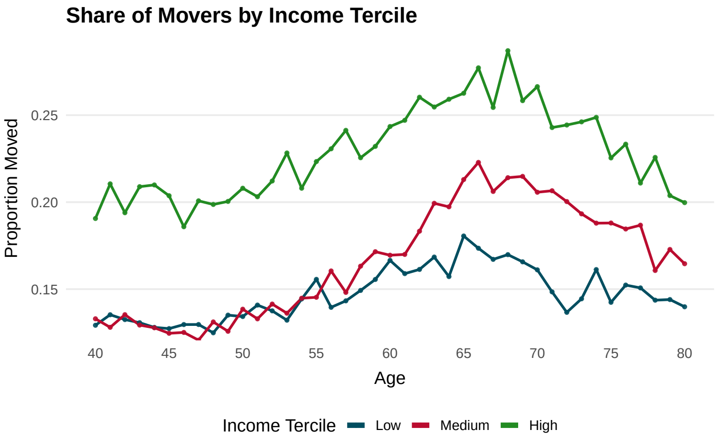

Figure: Income vs Moving Frequency

Graphical Evidence

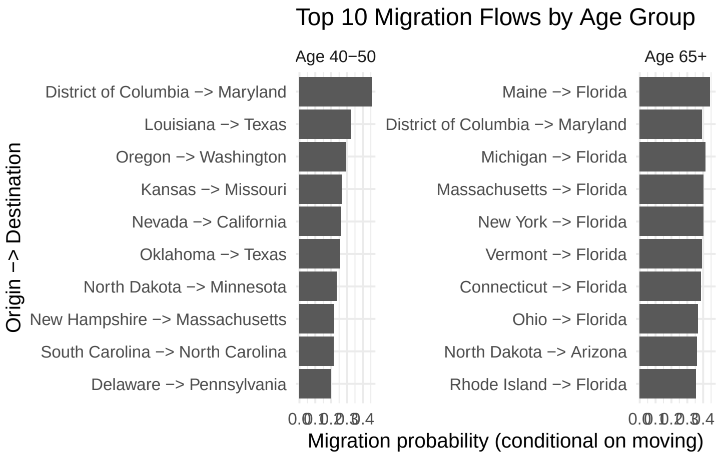

Figure: Top 10 Migration Flows

Model Overview

- Two types: Workers, Retirees
- Both types maximize lifetime utility over consumption and amenities
- Hold prior beliefs δ about all $z' \neq z$ which are discarded upon moving, and new beliefs are formed

Utility Function

- Workers: $U^w(c, ; t) = \frac{(c-\bar{c})^(1-\sigma)}{1-\sigma} - \gamma \cdot \{z' \neq z\}$
- Retirees: $U^r(c, m(z); t) = \frac{(c-\bar{c})^(1-\sigma)}{1-\sigma} + m(z) - \gamma \cdot \{z' \neq z\}$
- c: consumption, $m(z)$: amenities in location z
- γ: moving cost parameter

Budget Constraints

- Workers: $c = w(z) - \tau(z) - \mathbb{E}[c(h(z))|\delta]$
- δ: Prior assumptions about housing costs in location z'
- $w(z)$: wage in location z, $\tau(z)$: taxes, $c(h(z))$: housing cost (rent)
- Retirees: $c_t = p(1 - \tau(z))w(z) - \mathbb{E}[c(hc(z))|\delta]$
- p: pension replacement rate, $hc(z)$: health care cost

Decision Problem: Workers

$$V^w(\delta, z, t) = \max\{V_w^M(\delta, z, t), V_w^S(\delta, z, t)\} \ (1)$$

- V_w^M: value of moving, V_w^S: value of staying

$$V_w^S(\delta, z; t) = \max_c\{U(c) + V^w(\delta, z_t, t+1)\} \ (2)$$

$$V_w^M(\delta, z; t) = \max_{z', c}\{U(c) + \beta\mathbb{E}[V^w(\delta', z'; t+1)|\delta]|\delta])\} \ (3)$$

Decision Problem, Retirees

$$\mathrm{V}^r(\delta, z; t) = \max\{V_r^M(\delta, z; t), V_r^S(z; t)\} \ (4)$$

- V_r^M: value of moving, V_r^S: value of staying

$$V_r^S(z; t) = \max_c\{U(c, m(z)) + \beta V^r(\delta, z; t+1)\} \ (5)$$

$$V_r^M(\delta, z; t) = \max_{z', c}\{U(c, m(z')) + \beta V^r(\delta', z'; t+1)\} \ (6)$$

Estimation Strategy

- Use method of simulated moments (MSM)
- Set parameters: prior belief parameters δ, utility curvature σ, discount factor β
  - $\beta = 0.96$ fixed
  - $\sigma = 2$ fixed
  - $\delta = [-1, 0, 1]$
- Recover parameters: Moving cost parameter γ, δ for each group
- Simulated moments: age-specific moving rates and migration flow patterns

Graphical Results

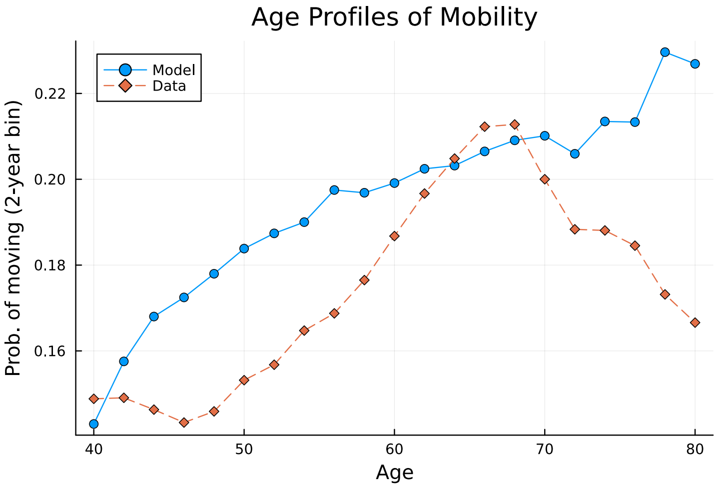

Figure: Age Profile of Moving Frequency: Model vs Data

Graphical Results

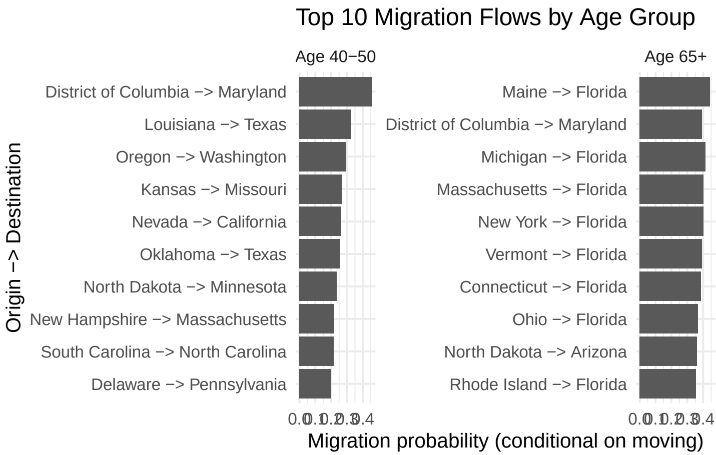

Figure: Migration Flows: Data

Graphical Results

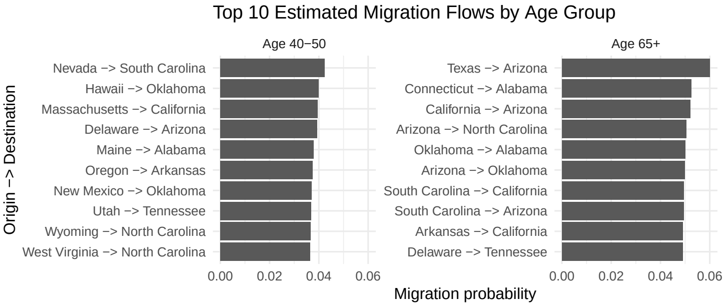

Figure: Migration Flows: Model

Discussion

- Model captures general trajectory of moving rates with age, with exception of oldest group
- Model fails to capture migration flow patterns for both groups
- Clearly more work to do on this front, suggestions welcome!

Thank You

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Questions?